

# MIAE 221: Materials Science for Engineers

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## Master Step-by-Step Quantitative Problem Guide

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Concordia University · Department of Mechanical, Industrial & Aerospace Engineering (MIAE)

**Standard:** Universal Step-by-Step Expansion Method (Practice Problems, Crystallography & Midterm Prep)

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### The Step-by-Step Expansion Methodology

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Following the universal standard established in [exact\\_ode\\_step\\_by\\_step.pdf](#) :

1. **Explicit Step Breakdown:** Every phase is numbered ( [Step 1](#) , [Step 2](#) , [Step 3](#) ...).
  2. **Theory Before Calculation:** The governing physical law, formula, or geometric theorem is defined before substituting values.
  3. **Zero Skipped Calculations:** All unit conversions, exponents, and intermediate products are fully displayed.
  4. **"Pattern to Remember" Algorithm:** Every problem ends with an algorithmic checklist to reproduce during exams.
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## Problem 1: Percent Ionic Character Calculation (Pauling Formula)

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### Problem Statement

Calculate the percent ionic character and percent covalent character of the chemical bond in Titanium Dioxide ( $\text{TiO}_2$ ).

Given Pauling electronegativity values

$X_{\text{Ti}} = 1.5$ ;  $X_{\text{O}} = 3.5$

$X_{\text{Ti}} = 1.5$ ;  $X_{\text{O}} = 3.5$ .

### Step 1: Identify electronegativities and state Pauling's formula

The degree of ionic character between two bonded elements  $A$  and  $B$  is quantified by Pauling's empirical equation:

where  $X_A$  and  $X_B$  are the respective Pauling electronegativities.

### Step 2: Compute the absolute electronegativity difference $\Delta X$

$$\Delta X = |X_{\text{O}} - X_{\text{Ti}}| = |3.5 - 1.5| = 2.0$$

### Step 3: Compute the squared difference and the exponent

$$(\Delta X)^2 = (2.0)^2 = 4.0$$

$$\text{Exponent} = -0.25 \times (\Delta X)^2 = -0.25 \times 4.0 = -1.0$$

### Step 4: Evaluate the exponential term

$$\exp(-1.0) = e^{-1} \approx 0.36788$$

### Step 5: Calculate the percent ionic and covalent character

Since total primary bonding character sums to 100:

**Conclusion:**  $\text{TiO}_2$  possesses a mixed bond that is predominantly **ionic**.

### ★ Pauling Ionic Character Pattern to Remember

1. Look up electronegativities  $X_A$  and  $X_B$ .
2. Compute  $\Delta X = |X_A - X_B|$ .
3. Calculate the exponent  $-0.25(\Delta X)^2$ .
4. Evaluate  $e^{\text{exponent}}$ .
5. Deduce % ionic character.

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## Problem 2: Equilibrium Interionic Spacing and Bonding Energy

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### Problem Statement

(Practice Problem Set #1 · Question 3)

For a potassium-chloride ( $K^+ - Cl^-$ ) ion pair, the attractive and repulsive potential energies vary with interionic separation  $r$

according to:

$$E_A(r) = -\frac{1.436}{r} \quad (\text{eV}), \quad E_R(r) = \frac{5.8 \times 10^{-6}}{r^9} \quad (\text{eV})$$

where  $r$  is expressed in nanometers (nm).

(a) Determine the equilibrium interionic separation  $r_0$ .

(b) Determine the equilibrium bonding energy  $E_0$ .

### Step 1: State the equilibrium thermodynamic condition

The net potential energy is the sum of attractive and repulsive energies:

$$E_N(r) = E_A(r) + E_R(r) = -1.436r^{-1} + 5.8 \times 10^{-6}r^{-9}$$

At the equilibrium distance  $r_0$ , the net interatomic force is zero ( $F_N = -dE_N/dr = 0$ ). Therefore, the potential energy is at a minimum:

$$\left. \frac{dE_N}{dr} \right|_{r=r_0} = 0$$

### Step 2: Differentiate $E_N(r)$ with respect to $r$

$$\begin{aligned} \frac{dE_N}{dr} &= \frac{d}{dr} [-1.436r^{-1} + 5.8 \times 10^{-6}r^{-9}] \\ &= (-1.436)(-1)r^{-2} + (5.8 \times 10^{-6})(-9)r^{-10} \\ \frac{dE_N}{dr} &= \frac{1.436}{r^2} - \frac{5.22 \times 10^{-5}}{r^{10}} \end{aligned}$$

### Step 3: Set the derivative equal to zero at $r = r_0$

$$\frac{1.436}{r_0^2} - \frac{5.22 \times 10^{-5}}{r_0^{10}} = 0 \implies \frac{1.436}{r_0^2} = \frac{5.22 \times 10^{-5}}{r_0^{10}}$$

### Step 4: Isolate and solve for $r_0$

Multiply both sides by  $r_0^{10}$ :

$$1.436r_0^8 = 5.22 \times 10^{-5}$$

Divide by 1.436:

$$r_0^8 = \frac{5.22 \times 10^{-5}}{1.436} = 3.6351 \times 10^{-5}$$

Take the 8th root:

$$r_0 = (3.6351 \times 10^{-5})^{1/8} = \mathbf{0.2789 \text{ nm}} \approx \mathbf{0.279 \text{ nm}} \quad (2.79 \text{ \AA})$$

### Step 5: Substitute $r_0$ into $E_N(r)$ to find the bonding energy $E_0$

$$E_0 = E_N(r_0) = -\frac{1.436}{0.2789} + \frac{5.8 \times 10^{-6}}{(0.2789)^9}$$

- Attractive term:  $-\frac{1.436}{0.2789} = -5.1488 \text{ eV}$
- Repulsive term:  $\frac{5.8 \times 10^{-6}}{(0.2789)^9} = \frac{5.8 \times 10^{-6}}{1.0138 \times 10^{-5}} = +0.5721 \text{ eV}$  Sum:

$$E_0 = -5.1488 + 0.5721 = \mathbf{-4.577 \text{ eV}} \approx \mathbf{-4.58 \text{ eV}}$$

## ★ Potential Well Pattern to Remember

1. Formulate  $E_N(r) = -Ar^{-1} + Br^{-n}$ .
2. Differentiate:  $\frac{dE_N}{dr} = Ar^{-2} - nBr^{-(n+1)}$ .
3. Set equal to 0 at  $r_0$  and isolate  $r_0^{n-1} = \frac{nB}{A}$ .
4. Solve for  $r_0 = \left(\frac{nB}{A}\right)^{1/(n-1)}$ .
5. Plug  $r_0$  back into  $E_N(r_0)$  to find bonding energy  $E_0$ .

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## Problem 3: Coulombic Electrostatic Force Balance Between Ions

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### Problem Statement

(Practice Problem Set #1 · Question 4)

Calculate the electrostatic attractive force and repulsive force between a  $K^+$  ion ( $z_1 = +1$ ) and an  $O^{2-}$  ion ( $z_2 = -2$ ) at an equilibrium spacing of  $r = 1.5$  nm.

### Step 1: State Coulomb's electrostatic force law

$$F_A = \frac{|z_1||z_2|e^2}{4\pi\epsilon_0 r^2}$$

### Step 2: List all physical constants and parameters in SI units

- Electronic charge:  $e = 1.602 \times 10^{-19}$  C
- Permittivity of free space:  $\epsilon_0 = 8.854 \times 10^{-12}$  C<sup>2</sup>/(N · m<sup>2</sup>)
- Valence magnitudes:  $|z_1| = 1, |z_2| = 2 \implies |z_1 z_2| = 2$
- Distance:  $r = 1.5$  nm =  $1.5 \times 10^{-9}$  m

### Step 3: Compute numerator and denominator separately

- Numerator:

$$|z_1 z_2|e^2 = 2 \times (1.602 \times 10^{-19})^2 = 2 \times 2.5664 \times 10^{-38} = 5.1328 \times 10^{-38} \text{ C}^2$$

- Denominator:

$$\begin{aligned} 4\pi\epsilon_0 r^2 &= 4\pi \times (8.854 \times 10^{-12}) \times (1.5 \times 10^{-9})^2 \\ &= 1.11265 \times 10^{-10} \times 2.25 \times 10^{-18} = 2.5035 \times 10^{-28} \text{ N} \cdot \text{m}^2 \end{aligned}$$

### Step 4: Compute the attractive force magnitude

$$F_A = \frac{5.1328 \times 10^{-38}}{2.5035 \times 10^{-28}} = 2.05 \times 10^{-10} \text{ N} \quad (0.205 \text{ nN})$$

### Step 5: Apply the equilibrium force balance condition

At the equilibrium distance  $r_0$ , the net force must be zero:

$$F_N = F_A + F_R = 0 \implies F_R = -F_A$$

Since attraction pulls inward ( $F_A = 2.05 \times 10^{-10}$  N), repulsion pushes outward with equal magnitude:

$$F_R = +2.05 \times 10^{-10} \text{ N} \quad (+0.205 \text{ nN})$$

### ★ Coulomb Force Pattern to Remember

1. Convert distance  $r$  from nm to meters ( $1 \text{ nm} = 10^{-9} \text{ m}$ ).
2. Use  $F_A = \frac{|z_1 z_2| e^2}{4\pi\epsilon_0 r^2}$ .
3. At equilibrium,  $F_R = -F_A$  (equal magnitude, opposite sign).

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## Problem 4: Wire Geometry to Microscopic Atom Count Stoichiometry

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### Problem Statement

(Practice Problem Set #1 · Question 5a)

Calculate the total number of gold (Au) atoms in a cylindrical gold wire with diameter  $d = 0.70 \text{ mm}$  and length  $L = 8.0 \text{ cm}$ .

Given

: Density of gold  $\rho = 19.3 \text{ g/cm}^3$ , atomic mass  $A_{Au} = 196.97 \text{ g/mol}$ , Avogadro's number  $N_A = 6.022 \times 10^{23} \text{ atoms/mol}$ .

### Step 1: Convert all dimensions to centimeters (cm)

- Diameter:  $d = 0.70 \text{ mm} = \frac{0.70}{10} \text{ cm} = 0.070 \text{ cm}$
- Radius:  $R = \frac{d}{2} = 0.035 \text{ cm}$
- Length:  $L = 8.0 \text{ cm}$

### Step 2: Compute geometric cross-sectional area and volume

- Area:  $A_{cross} = \pi R^2 = \pi (0.035 \text{ cm})^2 = 3.8485 \times 10^{-3} \text{ cm}^2$
- Volume:  $V = A_{cross} \times L = (3.8485 \times 10^{-3} \text{ cm}^2) \times 8.0 \text{ cm} = 0.030788 \text{ cm}^3$

### Step 3: Compute total mass from density

$$m = \rho \times V = 19.3 \frac{\text{g}}{\text{cm}^3} \times 0.030788 \text{ cm}^3 = 0.59421 \text{ g}$$

### Step 4: Compute number of moles

$$n_{mol} = \frac{m}{A_{Au}} = \frac{0.59421 \text{ g}}{196.97 \text{ g/mol}} = 3.0167 \times 10^{-3} \text{ mol}$$

### Step 5: Compute total number of atoms

$$N = n_{mol} \times N_A = (3.0167 \times 10^{-3} \text{ mol}) \times (6.022 \times 10^{23} \text{ atoms/mol})$$

$$\mathbf{N = 1.82 \times 10^{21} \text{ atoms}}$$

### ★ Macro-to-Micro Pattern to Remember

1. Convert all lengths to cm to match density units ( $\text{g/cm}^3$ ).
  2. Compute volume  $V = \frac{\pi}{4} d^2 L$ .
  3. Compute mass  $m = \rho V$ .
  4. Compute atoms  $N = \frac{m}{A} \times N_A$ .
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## Problem 5: Face-Centered Cubic (FCC) Lattice Parameter & APF Derivation

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### Problem Statement

Prove from first geometric principles that the Atomic Packing Factor (APF) of a Face-Centered Cubic (FCC) metallic crystal is  $\frac{\pi\sqrt{2}}{6} \approx 0.74$ .

### Step 1: Count the number of atoms per unit cell ( $n$ )

- 8 corner atoms, each shared among 8 adjacent unit cells:  $8 \times \frac{1}{8} = 1$  atom.
- 6 face-centered atoms, each shared between 2 adjacent unit cells:  $6 \times \frac{1}{2} = 3$  atoms.
- Total atoms per unit cell:

$$n = 1 + 3 = 4 \text{ atoms/cell}$$

### Step 2: Identify the close-packed direction and touching geometry

In FCC, atoms touch along the **face diagonals** of the cube (the family  $\langle 110 \rangle$ ). Across any cube face of side length  $a$ , the face diagonal contains:

- 1 radius from one corner atom:  $R$
- 1 full diameter from the central face atom:  $2R$
- 1 radius from the opposite corner atom:  $R$  Total diagonal length:

$$\text{Diagonal} = R + 2R + R = 4R$$

### Step 3: Relate lattice parameter $a$ to atomic radius $R$ via the Pythagorean Theorem

For a right triangle on the cube face with legs  $a$  and  $a$ :

$$a^2 + a^2 = (4R)^2 \implies 2a^2 = 16R^2 \implies a^2 = 8R^2$$

Taking the square root:

$$a = \sqrt{8}R = 2\sqrt{2}R$$

### Step 4: Calculate the volume of the unit cell ( $V_c$ )

$$V_c = a^3 = (2\sqrt{2}R)^3 = 2^3 \times (\sqrt{2})^3 \times R^3 = 8 \times 2\sqrt{2} \times R^3 = 16\sqrt{2}R^3$$

### Step 5: Calculate the total volume of atoms inside the unit cell ( $V_s$ )

Assuming hard spheres of radius  $R$ :

$$V_s = n \times \left( \frac{4}{3}\pi R^3 \right) = 4 \times \frac{4}{3}\pi R^3 = \frac{16}{3}\pi R^3$$

### Step 6: Compute the Atomic Packing Factor (APF)

$$\text{APF} = \frac{V_s}{V_c} = \frac{\frac{16}{3}\pi R^3}{16\sqrt{2}R^3} = \frac{16\pi}{3 \times 16\sqrt{2}} = \frac{\pi}{3\sqrt{2}}$$

Rationalize the denominator:  $\text{APF} = \frac{\pi}{3\sqrt{2}} = \frac{\pi\sqrt{2}}{3 \times 2} = \frac{\pi\sqrt{2}}{6} \approx 0.7405 \implies 0.74 \text{ (74\%)}$

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## Problem 6: Body-Centered Cubic (BCC) Lattice Parameter & APF Derivation

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### Problem Statement

Prove that for a Body-Centered Cubic (BCC) unit cell,  $a = \frac{4R}{\sqrt{3}}$  and  $\text{APF} = \frac{\pi\sqrt{3}}{8} \approx 0.68$ .

### Step 1: Count the number of atoms per unit cell ( $n$ )

- 8 corner atoms:  $8 \times \frac{1}{8} = 1$  atom.
- 1 central atom inside the body:  $1 \times 1 = 1$  atom.
- Total atoms per unit cell:

$$n = 1 + 1 = 2 \text{ atoms/cell}$$

### Step 2: Identify the close-packed direction

In BCC, atoms touch along the **body diagonals** of the cube (family  $\langle 111 \rangle$ ). Along the 3D body diagonal:

$$\text{Body Diagonal} = R + 2R + R = 4R$$

### Step 3: Relate $a$ to $R$ using 3D Pythagorean Theorem

The body diagonal of a cube with edge  $a$  satisfies:

$$a^2 + a^2 + a^2 = (4R)^2 \implies 3a^2 = 16R^2 \implies a\sqrt{3} = 4R$$

$$a = \frac{4R}{\sqrt{3}}$$

### Step 4: Calculate unit cell volume ( $V_c$ )

$$V_c = a^3 = \left(\frac{4R}{\sqrt{3}}\right)^3 = \frac{64R^3}{3\sqrt{3}}$$

### Step 5: Calculate atomic volume ( $V_s$ )

$$V_s = n \times \left(\frac{4}{3}\pi R^3\right) = 2 \times \frac{4}{3}\pi R^3 = \frac{8}{3}\pi R^3$$

### Step 6: Compute APF

$$\text{APF} = \frac{V_s}{V_c} = \frac{\frac{8}{3}\pi R^3}{\frac{64R^3}{3\sqrt{3}}} = \frac{8\pi}{64} \times \sqrt{3} = \frac{\pi\sqrt{3}}{8} \approx 0.6802 \implies 0.68 \text{ (68\%)} \quad \blacksquare$$

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## Problem 7: Theoretical Density Computation of FCC Copper

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### Problem Statement

(Lecture 5 · Example Problem)

Calculate the theoretical density of copper (Cu), given:

- Atomic radius:  $R = 0.128$  nm
- Crystal structure: Face-Centered Cubic (FCC)
- Atomic mass:  $A_{Cu} = 63.55$  g/mol

- Avogadro's number:  $N_A = 6.022 \times 10^{23}$  atoms/mol

### Step 1: Identify parameters from crystal structure

For FCC:

- Number of atoms per unit cell:  $n = 4$
- Lattice parameter relation:  $a = 2\sqrt{2}R$

### Step 2: Calculate lattice parameter $a$ in centimeters (cm)

$$a = 2\sqrt{2}(0.128 \text{ nm}) = 2.8284 \times 0.128 \text{ nm} = 0.36204 \text{ nm}$$

Convert from nm to cm ( $1 \text{ nm} = 10^{-7} \text{ cm}$ ):

$$a = 0.36204 \times 10^{-7} \text{ cm} = 3.6204 \times 10^{-8} \text{ cm}$$

### Step 3: Calculate unit cell volume $V_c$ in $\text{cm}^3$

$$V_c = a^3 = (3.6204 \times 10^{-8} \text{ cm})^3 = 4.7454 \times 10^{-23} \text{ cm}^3$$

### Step 4: State the theoretical density formula

$$\rho = \frac{n \cdot A}{V_c \cdot N_A}$$

### Step 5: Substitute all values and compute $\rho$

- Mass of unit cell:

$$n \cdot A = 4 \times 63.55 \text{ g/mol} = 254.2 \text{ g/mol}$$

- Denominator ( $V_c \cdot N_A$ ):

$$V_c \cdot N_A = (4.7454 \times 10^{-23} \text{ cm}^3) \times (6.022 \times 10^{23} \text{ mol}^{-1}) = 28.577 \text{ cm}^3/\text{mol}$$

- Density:

$$\rho = \frac{254.2}{28.577} = 8.895 \text{ g/cm}^3 \approx 8.90 \text{ g/cm}^3$$

(This theoretical value closely matches the experimental literature density of Copper:  $8.94 \text{ g/cm}^3$ , confirming accuracy).

## Problem 8: Crystallographic Direction Miller Indices $[uvw]$

### Problem Statement

Determine the Miller indices for the direction vector starting at coordinates  $(0, 0, 1)$  and terminating at coordinates  $(1, 1, 0)$ .

### Step 1: Identify coordinates of Tail and Head

- **Tail:**  $(x_1, y_1, z_1) = (0, 0, 1)$
- **Head:**  $(x_2, y_2, z_2) = (1, 1, 0)$

## Step 2: Compute Head minus Tail differences

$$\Delta x = x_2 - x_1 = 1 - 0 = 1$$

$$\Delta y = y_2 - y_1 = 1 - 0 = 1$$

$$\Delta z = z_2 - z_1 = 0 - 1 = -1$$

## Step 3: Clear fractions and reduce to smallest integers

The vector components are already integers:  $(1, 1, -1)$ .

## Step 4: Format with square brackets and overbars

Enclose in square brackets with no commas; replace  $-1$  with an overbar  $\bar{1}$ :

$$[11\bar{1}]$$

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## Problem 9: Crystallographic Plane Miller Indices $(hkl)$

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### Problem Statement

Determine the Miller indices  $(hkl)$  of a crystallographic plane that intersects the  $x$ -axis at  $x = a$ , the  $y$ -axis at  $y = \frac{2}{3}b$ , and is parallel to the  $z$ -axis.

### Step 1: Origin check

The plane does not pass through the coordinate origin  $(0, 0, 0)$ . No origin shift is required.

### Step 2: Determine axial intercepts in units of $a, b, c$

- $x$ -intercept: 1
- $y$ -intercept:  $\frac{2}{3}$
- $z$ -intercept:  $\infty$  (since the plane is parallel to the  $z$ -axis)

### Step 3: Take reciprocals of the intercepts

- $h = \frac{1}{1} = 1$
- $k = \frac{1}{2/3} = \frac{3}{2}$
- $l = \frac{1}{\infty} = 0$

### Step 4: Clear fractions by multiplying by the lowest common denominator

Multiply all three terms by 2:

- $h = 1 \times 2 = 2$
- $k = \frac{3}{2} \times 2 = 3$
- $l = 0 \times 2 = 0$

### Step 5: Enclose in parentheses

$$(230)$$

*(The family of all symmetrically equivalent planes in cubic crystals is denoted with braces:  $\{230\}$ ).*

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## Problem 10: Linear Density of Crystallographic Directions

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### Problem Statement

(Lecture 6 - Slide 4 & Callister §3.11)

- (a) Derive the expression for the **Linear Density** ( $LD$ ) of the  $[100]$  direction in an FCC unit cell in terms of the atomic radius  $R$ .  
(b) Calculate the numerical linear density for Copper ( $R = 0.128 \text{ nm}$ ).  
(c) Derive  $LD$  for the close-packed  $[110]$  direction in FCC and demonstrate that it attains the theoretical maximum packing of  $LD = \frac{1}{2R}$ .
- 

### Part (a): Linear Density of $[100]$ in FCC

#### Step 1: State the governing equation for Linear Density

$$LD = \frac{\text{Number of atomic diameters along direction vector inside unit cell}}{\text{Length of the direction vector inside unit cell}} = \frac{n}{L_L}$$

#### Step 2: Count atoms centered on the $[100]$ vector segment

The  $[100]$  direction vector runs along the bottom edge of the cubic unit cell from  $(0, 0, 0)$  to  $(1, 0, 0)$ .

- It passes through two corner atoms.
- Each corner atom is centered at the vertex, contributing only  $\frac{1}{2}$  of an atom to this line segment:

$$n = 2 \times \frac{1}{2} = 1 \text{ atom}$$

#### Step 3: Determine the line length $L_L$

The vector length is equal to one lattice parameter  $a$ . For FCC, atoms touch along the face diagonal ( $a\sqrt{2} = 4R$ ):

$$L_L = a = 2\sqrt{2}R$$

#### Step 4: Compute Linear Density

$$LD_{[100]} = \frac{n}{L_L} = \frac{1}{2\sqrt{2}R} = \frac{\sqrt{2}}{4R} \approx \frac{\mathbf{0.3536}}{\mathbf{R}}$$

### Part (b): Numerical calculation for Copper

Given  $R_{\text{Cu}} = 0.128 \text{ nm} = 0.128 \times 10^{-9} \text{ m}$ :

$$LD_{[100]} = \frac{1}{2\sqrt{2}(0.128 \text{ nm})} = \frac{1}{0.36204 \text{ nm}} = \mathbf{2.762 \text{ atoms/nm}} = \mathbf{2.762 \times 10^9 \text{ atoms/m}}$$

### Part (c): Linear Density of Close-Packed $[110]$ in FCC

- The  $[110]$  vector runs diagonally across the cube face from  $(0, 0, 0)$  to  $(1, 1, 0)$ .

- It passes through two corner atoms (each contributing  $\frac{1}{2}$ ) and one full face-centered atom:

$$n = 2 \left( \frac{1}{2} \right) + 1 = \mathbf{2 \text{ atoms}}$$

- The length of the face diagonal is:

$$L_L = a\sqrt{2} = (2\sqrt{2}R)\sqrt{2} = 4R$$

- Therefore:

$$\text{LD}_{[110]} = \frac{2}{4R} = \frac{1}{2R}$$

- For Copper:

$$\text{LD}_{[110]} = \frac{1}{2(0.128 \text{ nm})} = \mathbf{3.906 \text{ atoms/nm}} = \mathbf{3.906 \times 10^9 \text{ atoms/m}}$$

- *Significance*  
: Since atoms touch continuously along  $[110]$  ( $2R$  per atom), this is the close-packed direction where linear density is maximized!

## Problem 11: Planar Density of Crystallographic Planes

### Problem Statement

(Lecture 6 · Slide 6 & Callister §3.11)

- Derive the **Planar Density** ( $PD$ ) of the  $(110)$  plane in an FCC crystal in terms of  $R$ .
- Derive the Planar Density of the close-packed  $(111)$  plane in an FCC crystal.
- For Aluminum ( $R = 0.143 \text{ nm}$ ), compute the numerical planar density of  $(111)$  in  $\text{atoms/nm}^2$ .

### Part (a): Planar Density of FCC $(110)$

#### Step 1: State the governing equation for Planar Density

$$PD = \frac{\text{Number of atoms centered on plane inside unit cell}}{\text{Area of the plane inside unit cell}} = \frac{n_P}{A_P}$$

#### Step 2: Count atoms centered on the $(110)$ plane

The  $(110)$  plane cuts diagonally through the FCC cube, forming a rectangle of dimensions  $a \times a\sqrt{2}$ .

- 4 corner atoms: each shared by 4 adjacent unit cells on this plane  $\rightarrow 4 \times \frac{1}{4} = 1 \text{ atom}$ .
- 2 face-centered atoms (top and bottom faces): cut in half by this plane  $\rightarrow 2 \times \frac{1}{2} = 1 \text{ atom}$ .
- Total atoms centered on the plane:

$$n_P = 4 \left( \frac{1}{4} \right) + 2 \left( \frac{1}{2} \right) = 1 + 1 = \mathbf{2 \text{ atoms}}$$

### Step 3: Calculate the area of the (110) rectangle

$$A_P = a \times a\sqrt{2} = \sqrt{2} a^2$$

Substitute  $a = 2\sqrt{2}R$ :

$$A_P = \sqrt{2} (2\sqrt{2}R)^2 = \sqrt{2} (8R^2) = 8\sqrt{2} R^2$$

### Step 4: Compute Planar Density

$$PD_{(110)} = \frac{2}{8\sqrt{2} R^2} = \frac{1}{4\sqrt{2} R^2} = \frac{\sqrt{2}}{8R^2} \approx \frac{0.1768}{R^2}$$

### Part (b): Planar Density of Close-Packed FCC (111)

- The (111) plane passes through three face diagonals, forming an **equilateral triangle** with side length  $s = a\sqrt{2} = 4R$ .
- Atom count on the (111) triangle:
  - 3 corner atoms, each interior angle is  $60^\circ$  (contributes  $\frac{60^\circ}{360^\circ} = \frac{1}{6}$ ):  $3 \times \frac{1}{6} = \frac{1}{2}$ .
  - 3 face-center atoms along the edges, each shared between 2 adjacent unit cell planes (contributes  $\frac{1}{2}$ ):  $3 \times \frac{1}{2} = \frac{3}{2}$ .
  - Total atoms:

$$n_P = \frac{1}{2} + \frac{3}{2} = 2 \text{ atoms}$$

- Area of the equilateral triangle:

$$A_P = \frac{\sqrt{3}}{4} s^2 = \frac{\sqrt{3}}{4} (4R)^2 = 4\sqrt{3} R^2$$

- Planar Density:

$$PD_{(111)} = \frac{2}{4\sqrt{3} R^2} = \frac{1}{2\sqrt{3} R^2} = \frac{\sqrt{3}}{6R^2} \approx \frac{0.2887}{R^2}$$

### Part (c): Numerical calculation for Aluminum

Given  $R_{Al} = 0.143 \text{ nm}$ :

$$PD_{(111)} = \frac{1}{2\sqrt{3} (0.143 \text{ nm})^2} = \frac{1}{3.4641 \times 0.020449} = \frac{1}{0.070838 \text{ nm}^2} = 14.12 \text{ atoms/nm}^2 = 1.412 \times 10^{19} \text{ atoms/m}^2$$

## Problem 12: X-Ray Diffraction, Interplanar Spacing & Bragg's Law

### Problem Statement

(Lecture 6 · Slide 14–17 & Callister §3.16)

Diffraction angles are measured for FCC Copper ( $a = 0.3615 \text{ nm}$ ) using monochromatic X-radiation with wavelength  $\lambda = 0.1542 \text{ nm}$  (Cu  $K_\alpha$ ).

- Calculate the interplanar spacing  $d_{hkl}$  for the (111) and (200) planes.
- Determine the diffraction angle ( $2\theta$ ) for first-order ( $n = 1$ ) reflection from the (111) plane.
- State the diffraction selection rules for FCC and BCC crystals.

### Step 1: Calculate interplanar spacing $d_{hkl}$

For cubic crystal systems:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

- For (111) plane:

$$d_{111} = \frac{0.3615}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{0.3615}{\sqrt{3}} = \frac{0.3615}{1.73205} = \mathbf{0.2087 \text{ nm}} \quad (2.087 \text{ \AA})$$

- For (200) plane:

$$d_{200} = \frac{0.3615}{\sqrt{2^2 + 0^2 + 0^2}} = \frac{0.3615}{\sqrt{4}} = \frac{0.3615}{2} = \mathbf{0.1808 \text{ nm}} \quad (1.808 \text{ \AA})$$

### Step 2: Apply Bragg's Law to find $\theta$ and $2\theta$

Bragg's Law governs constructive interference of X-rays:

$$n\lambda = 2d_{hkl} \sin \theta$$

For first-order diffraction ( $n = 1$ ):

$$\sin \theta = \frac{n\lambda}{2d_{111}} = \frac{1 \times 0.1542 \text{ nm}}{2 \times 0.2087 \text{ nm}} = \frac{0.1542}{0.4174} = 0.36943$$

Take the inverse sine:

$$\theta = \arcsin(0.36943) = \mathbf{21.68^\circ}$$

The detector angle recorded on a diffractometer is  $2\theta$ :

$$\mathbf{2\theta = 2 \times 21.68^\circ = 43.36^\circ}$$

### Step 3: Diffraction Reflection Selection Rules

Not all planes produce diffraction peaks due to destructive interference of waves scattered by interior atoms:

| Crystal System | Allowed Reflection Rule                             | First 6 Diffraction Peaks (Lowest to Highest $2\theta$ ) |
|----------------|-----------------------------------------------------|----------------------------------------------------------|
| BCC            | $h + k + l = \text{even integer}$                   | (110), (200), (211), (220), (310), (222)                 |
| FCC            | $h, k, l$ must be <b>all odd</b> OR <b>all even</b> | (111), (200), (220), (311), (222), (400)                 |

(Notice that for FCC, (111) is the first peak because 1, 1, 1 are all odd; (200) is second because 2, 0, 0 are all even; (100) is forbidden because 1, 0, 0 is mixed).

## Problem 13: Allotropic / Polymorphic Volume Change

### Problem Statement

(Lecture 6 · Slide 11 & Callister §3.10)

Pure Iron undergoes an allotropic phase transformation upon heating through  $912^\circ\text{C}$  from  $\alpha$ -iron (BCC,  $a_\alpha = 0.2866 \text{ nm}$ ) to  $\gamma$ -iron

(FCC,  $a_\gamma = 0.3571 \text{ nm}$ ).

Calculate the percentage volume change ( $\Delta V/V_\alpha \times 100$ ) that accompanies this transformation. State whether iron expands or contracts upon heating through  $912^\circ\text{C}$ .

---

### Step 1: Relate crystal volume to atomic volume

Mass is conserved during phase transformation. Therefore, the volume comparison must be made **per individual atom** (or per mole):

$$V_{\text{atom}} = \frac{V_{\text{unit cell}}}{n_{\text{atoms/cell}}} = \frac{a^3}{n}$$

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### Step 2: Compute volume per atom in $\alpha$ -iron (BCC)

In BCC,  $n = 2$  atoms/cell:

$$V_{\text{cell}, \alpha} = a_\alpha^3 = (0.2866 \text{ nm})^3 = 0.023542 \text{ nm}^3$$
$$V_{\text{atom}, \alpha} = \frac{0.023542 \text{ nm}^3}{2} = \mathbf{0.011771 \text{ nm}^3/\text{atom}}$$

---

### Step 3: Compute volume per atom in $\gamma$ -iron (FCC)

In FCC,  $n = 4$  atoms/cell:

$$V_{\text{cell}, \gamma} = a_\gamma^3 = (0.3571 \text{ nm})^3 = 0.045538 \text{ nm}^3$$
$$V_{\text{atom}, \gamma} = \frac{0.045538 \text{ nm}^3}{4} = \mathbf{0.011385 \text{ nm}^3/\text{atom}}$$

---

### Step 4: Calculate percentage volume change

(Using high-temperature thermal dilation values at exactly  $912^\circ\text{C}$ ,  $a_\alpha = 0.2892 \text{ nm}$  and  $a_\gamma = 0.3643 \text{ nm}$ , yielding  $\approx -1.2$ ).

#### Physical Conclusion:

**Iron CONTRACTS upon heating from BCC to FCC!**

*Physical Reason*

: FCC has an atomic packing factor of  $\text{APF} = 0.74$ , whereas BCC has  $\text{APF} = 0.68$ . Transforming into a closer-packed structure packs the atoms more tightly, causing a net volumetric shrinkage despite the increase in temperature.

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## Problem 14: Equilibrium Vacancy Concentration & Arrhenius Activation Energy

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### Problem Statement

(Lecture 7 · Slide 4 & Callister §4.2)

Calculate the equilibrium number of vacancies per cubic meter ( $N_v$ ) in pure Copper at  $1000^\circ\text{C}$  (1273 K).

*Given*

:

- Energy for vacancy formation:  $Q_v = 0.90 \text{ eV/atom}$
- Boltzmann's constant:  $k = 8.62 \times 10^{-5} \text{ eV/K}$
- Density of Copper at  $1000^\circ\text{C}$ :  $\rho = 8.40 \text{ g/cm}^3 = 8.40 \times 10^6 \text{ g/m}^3$
- Atomic weight of Copper:  $A_{\text{Cu}} = 63.55 \text{ g/mol}$
- Avogadro's number:  $N_A = 6.022 \times 10^{23} \text{ atoms/mol}$

### Step 1: Calculate total number of atomic lattice sites per cubic meter ( $N$ )

$$N = \frac{\rho \cdot N_A}{A_{\text{Cu}}}$$

$$N = \frac{(8.40 \times 10^6 \text{ g/m}^3) \times (6.022 \times 10^{23} \text{ atoms/mol})}{63.55 \text{ g/mol}} = \frac{5.0585 \times 10^{30}}{63.55} = \mathbf{7.960 \times 10^{28} \text{ sites/m}^3}$$

### Step 2: State the Arrhenius vacancy equation

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right)$$

### Step 3: Compute the thermal activation denominator ( $kT$ )

$$T = 1000 + 273 = 1273 \text{ K}$$

$$kT = (8.62 \times 10^{-5} \text{ eV/K}) \times (1273 \text{ K}) = \mathbf{0.10973 \text{ eV}}$$

### Step 4: Evaluate the exponential Boltzmann factor

$$\text{Exponent} = -\frac{Q_v}{kT} = -\frac{0.90 \text{ eV}}{0.10973 \text{ eV}} = -8.2019$$

$$\exp(-8.2019) = e^{-8.2019} = \mathbf{2.741 \times 10^{-4}}$$

### Step 5: Compute the equilibrium vacancy concentration $N_v$

$$N_v = (7.960 \times 10^{28} \text{ sites/m}^3) \times (2.741 \times 10^{-4}) = \mathbf{2.182 \times 10^{25} \text{ vacancies/m}^3}$$

**Vacancy Fraction:**

$$\frac{N_v}{N} = 2.741 \times 10^{-4} = \frac{1}{3648}$$

(At  $1000^\circ\text{C}$ , approximately 1 out of every 3,650 lattice sites in copper is vacant! At room temperature  $25^\circ\text{C}$ ,  $N_v/N \approx 10^{-15}$ , demonstrating that vacancy concentration increases exponentially with temperature).

## Problem 15: Hume-Rothery Quantitative Solid Solubility Evaluation

### Problem Statement

(Lecture 7 · Slide 13 & Callister §4.3)

Using the Hume-Rothery rules, evaluate whether Zinc (Zn) can form an unlimited/complete substitutional solid solution in Copper (Cu). Compare this with the Copper-Nickel (Cu-Ni) system.

| Element     | Atomic Radius ( $R$ ) | Crystal Structure | Electronegativity ( $X$ ) | Valence |
|-------------|-----------------------|-------------------|---------------------------|---------|
| Copper (Cu) | 0.128 nm              | FCC               | 1.9                       | +2      |
| Zinc (Zn)   | 0.133 nm              | HCP               | 1.6                       | +2      |
| Nickel (Ni) | 0.125 nm              | FCC               | 1.8                       | +2      |

### Step 1: Rule 1 — Atomic Size Factor (Difference $\leq 15$ )

$$\Delta R = \left| \frac{R_{\text{solute}} - R_{\text{solvent}}}{R_{\text{solvent}}} \right| \times 100$$

- For Cu-Zn:

$$\Delta R = \left| \frac{0.133 - 0.128}{0.128} \right| \times 100$$

- For Cu-Ni:

$$\Delta R = \left| \frac{0.125 - 0.128}{0.128} \right| \times 100$$

### Step 2: Rule 2 — Crystal Structure Rule (Must be IDENTICAL)

- For Cu-Zn: Cu is **FCC** while Zn is **HCP**. **[VIOLATED]**

*Consequence*

: Because the crystal structures are different, Zn CANNOT dissolve completely in Cu. It forms only a partial solid solution (maximum solubility of  $\approx 35$  wt at room temperature, forming  $\alpha$ -brass; beyond that, a new phase  $\beta$  forms).

- For Cu-Ni: Cu is **FCC** and Ni is **FCC**. **[SATISFIED]**

### Step 3: Rule 3 — Electronegativity Difference ( $|\Delta X| \leq 0.4$ )

- For Cu-Zn:  $|\Delta X| = |1.6 - 1.9| = 0.3 \leq 0.4$  **[SATISFIED]**
- For Cu-Ni:  $|\Delta X| = |1.8 - 1.9| = 0.1 \leq 0.4$  **[SATISFIED]**

### Step 4: Rule 4 — Valency

- Both pairs have identical primary valency of +2. **[SATISFIED]**

### Summary Table & Exam Takeaway:

- Cu-Ni **System**: Satisfies all 4 Hume-Rothery rules  $\implies$  **Complete 100 Isomorphous Solid Solubility** across all compositions from 0 to 100 Ni.
- Cu-Zn **System**: Violates the crystal structure rule  $\implies$  **Partial Solid Solubility** only (limited to  $\approx 35$  wt Zn).

## Problem 16: Composition Conversions Between Weight Percent and Atom Percent

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### Problem Statement

(Lecture 7 · Slide 12 & Callister §4.4)

- (a) A cartridge brass alloy consists of 70.0 wt Copper ( $\text{Cu}$ ,  $A_{\text{Cu}} = 63.55 \text{ g/mol}$ ) and 30.0 wt Zinc ( $\text{Zn}$ ,  $A_{\text{Zn}} = 65.38 \text{ g/mol}$ ). Convert this composition to **atom percent** (*at*) of Zinc.
- (b) Convert an alloy with 15.0 at Silicon ( $A_{\text{Si}} = 28.09 \text{ g/mol}$ ) in Aluminum ( $A_{\text{Al}} = 26.98 \text{ g/mol}$ ) to **weight percent** (*wt*).
- 

### Part (a): Weight Percent to Atom Percent

**Method:** 100 g Basis

Assume a total sample mass of 100 g:

- Mass of Cu:  $m_{\text{Cu}} = 70.0 \text{ g}$
- Mass of Zn:  $m_{\text{Zn}} = 30.0 \text{ g}$

Compute number of moles of each element:

$$n_{\text{Cu}} = \frac{m_{\text{Cu}}}{A_{\text{Cu}}} = \frac{70.0 \text{ g}}{63.55 \text{ g/mol}} = 1.1015 \text{ mol}$$
$$n_{\text{Zn}} = \frac{m_{\text{Zn}}}{A_{\text{Zn}}} = \frac{30.0 \text{ g}}{65.38 \text{ g/mol}} = 0.4589 \text{ mol}$$

Total moles:

$$n_{\text{total}} = 1.1015 + 0.4589 = 1.5604 \text{ mol}$$

Compute atom percent (*at*):

$$C'_{\text{Zn}} = \frac{n_{\text{Zn}}}{n_{\text{total}}} \times 100$$
$$C'_{\text{Cu}} = 100$$

---

### Part (b): Atom Percent to Weight Percent

**Method:** 100 mol Basis

Assume a total of 100 moles of alloy:

- Moles of Si:  $n_{\text{Si}} = 15.0 \text{ mol}$
- Moles of Al:  $n_{\text{Al}} = 85.0 \text{ mol}$

Compute mass of each element:

$$m_{\text{Si}} = n_{\text{Si}} \times A_{\text{Si}} = 15.0 \text{ mol} \times 28.09 \text{ g/mol} = 421.35 \text{ g}$$
$$m_{\text{Al}} = n_{\text{Al}} \times A_{\text{Al}} = 85.0 \text{ mol} \times 26.98 \text{ g/mol} = 2293.30 \text{ g}$$

Total mass:

$$m_{\text{total}} = 421.35 + 2293.30 = 2714.65 \text{ g}$$

Compute weight percent (*wt*):

$$C_{\text{Si}} = \frac{m_{\text{Si}}}{m_{\text{total}}} \times 100$$
$$C_{\text{Al}} = 100$$

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## Problem 17: Dislocation Burgers Vector Magnitude in FCC and BCC Slip Systems

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### Problem Statement

(Lecture 7 · Slide 18–20 & Callister §4.5)

Dislocations slip along close-packed directions, which define the orientation and magnitude of the **Burgers vector**  $\vec{b}$ .

(a) In FCC metals, the Burgers vector is of the type  $\vec{b} = \frac{a}{2}\langle 110 \rangle$ . Compute the magnitude  $|\vec{b}|$  for Copper ( $a = 0.3615 \text{ nm}$ ). Show that  $|\vec{b}|$  equals exactly one atomic diameter ( $2R$ ).

(b) In BCC metals, the Burgers vector is of the type  $\vec{b} = \frac{a}{2}\langle 111 \rangle$ . Compute the magnitude  $|\vec{b}|$  for  $\alpha$ -Iron ( $a = 0.2866 \text{ nm}$ ).

---

### Part (a): Burgers Vector Magnitude in FCC

The Burgers vector is  $\vec{b} = \frac{a}{2}[110]$ :

$$|\vec{b}| = \frac{a}{2} \sqrt{1^2 + 1^2 + 0^2} = \frac{a\sqrt{2}}{2} = \frac{a}{\sqrt{2}}$$

Substitute  $a_{\text{Cu}} = 0.3615 \text{ nm}$ :

$$|\vec{b}_{\text{Cu}}| = \frac{0.3615 \text{ nm}}{\sqrt{2}} = \frac{0.3615}{1.4142} = \mathbf{0.2556 \text{ nm}} \quad (2.556 \text{ \AA})$$

**Proof that  $|\vec{b}| = 2R$ :**

In FCC, atoms touch along the face diagonal:  $a = 2\sqrt{2}R$ .

$$|\vec{b}| = \frac{a}{\sqrt{2}} = \frac{2\sqrt{2}R}{\sqrt{2}} = \mathbf{2R}$$

*Interpretation*

: When a dislocation glides through an FCC crystal, the lattice is displaced by exactly one full atomic diameter!

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### Part (b): Burgers Vector Magnitude in BCC

The Burgers vector is  $\vec{b} = \frac{a}{2}[111]$ :

$$|\vec{b}| = \frac{a}{2} \sqrt{1^2 + 1^2 + 1^2} = \frac{a\sqrt{3}}{2}$$

Substitute  $a_{\text{Fe}} = 0.2866 \text{ nm}$ :

$$|\vec{b}_{\text{Fe}}| = \frac{0.2866 \times 1.73205}{2} = \frac{0.4964}{2} = \mathbf{0.2482 \text{ nm}} \quad (2.482 \text{ \AA})$$

**Proof that  $|\vec{b}| = 2R$ :**

In BCC, atoms touch along the body diagonal:  $a = \frac{4R}{\sqrt{3}}$ .

$$|\vec{b}| = \frac{a\sqrt{3}}{2} = \frac{\frac{4R}{\sqrt{3}} \cdot \sqrt{3}}{2} = \frac{4R}{2} = \mathbf{2R}$$

*Interpretation*

: In both FCC and BCC systems, the Burgers vector magnitude corresponds precisely to the interatomic touching distance  $2R$  along the primary slip direction!